Fiscal Year:	FY 2004	**Task Last Updated:**	FY 06/26/2009
PI Name:	Rajulu, Sudhakar Ph.D.		
Project Title:	Development of Analytical Tools to Process and Apply Digitally Scanned Anthropometric Data		
Division Name:	Human Research		
Program/Discipline:	ADVANCED HUMAN SUPPORT TECHNOLOGIES		
Program/Discipline--Element/Subdiscipline:	ADVANCED HUMAN SUPPORT TECHNOLOGIES--Space human factors engineering		
Joint Agency Name:		**TechPort:**	No
Human Research Program Elements:	None		
Human Research Program Risks:	None		
Space Biology Element:	None		
Space Biology Cross-Element Discipline:	None		
Space Biology Special Category:	None		
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PI Organization Type:	NASA CENTER	**Phone:**	281-483-3725
Organization Name:	NASA Johnson Space Center		
PI Address 1:	Code SF3		
PI Address 2:	2101 NASA Pkwy		
PI Web Page:			
City:	Houston	**State:**	TX
Zip Code:	77058	**Congressional District:**	22
Comments:			
Project Type:	Ground	**Solicitation / Funding Source:**	INTERNAL PROJECT
Start Date:	08/01/2001	**End Date:**	08/01/2004
No. of Post Docs:	2	**No. of PhD Degrees:**	
No. of PhD Candidates:	0	**No. of Master' Degrees:**	
No. of Master's Candidates:	0	**No. of Bachelor's Degrees:**	
No. of Bachelor's Candidates:	3	**Monitoring Center:**	NASA JSC
Contact Monitor:		**Contact Phone:**	
Contact Email:			
Flight Program:			
Flight Assignment:			
Key Personnel Changes/Previous PI:			
COI Name (Institution):	Gonzalez, Luis (Lockheed Martin)		
Grant/Contract No.:			
Performance Goal No.:			
Performance Goal Text:			
Task Description:	The proposed project type research is to enhance the core capabilities of the Habitability and Human Factors office at the Johnson Space Center in the area of gathering and utilizing 3-dimensional whole body anthropometric data. This capability is very much in need to improve the design of space suits and other space hardware in such a way to enhance crew comfort and safety. The objectives of this project are threefold. First, develop analytical software tools to process whole-body scanned data. Second, verify, modify, and validate the analytical software results. Third, develop statistical analysis software to perform Gap analysis on existing space hardware. The deliverables will include analytical software tools to extract anthropometric data (linear, curvilinear, and volumetric) from the whole-body scans. In addition, several statistical analysis codes will be developed to perform Gap analyses of all existing space hardware.		

Rationale for HRP Directed Research:	
Research Impact/Earth Benefits:	The methodolgies and data extraction methods and the extracted data themselves will be of immense use for those engaged in computer modeling and simulation of humans interacting with a work environment.
Task Progress:	• Linear measurements o A study to validate linear measurements has been completed. • Develop method to merge scan o A method to merge scans has been developed and documented. • Develop method to fill voids o A method to fill the voids in scans has been developed and documented. • Circumferential measurements o Circumferential measurements have been validated. • Develop method to compute volume, mass and area of scanned segments o A software program named ‘Algor’ is being used in conjunction with the ‘ScanWorx’ and ‘Polyworks’ software to develop a method to compute the volume, area and mass properties of scanned segments. • Protocol for seated scans o A chair for the seated scans is being designed and the protocol is being developed. • Animate segmented scans o Current Vicon model is being studied to see how it can be used as the model to drive the animation.
Bibliography Type:	Description: (Last Updated: 03/25/2020)